

one-half. This makes the variance estimates less stable (i.e., the variance of the variance estimates is increased).

Variance estimation from complex samples such as NHANES III is a complicated process. Because of the estimation techniques used, exact expressions for sampling errors are often not available and approximations must be used. Also, estimates of sampling error are themselves subject to variability. Thus, analysts and users should be careful in analyzing data from complex samples and drawing inferences. It is tempting to treat survey data as if it were derived from a simple random sample; if this were the case then standard methods and software could be employed. However, it must be emphasized that standard statistical analyses based on simple random sampling are generally NOT directly applicable to complex samples.

If topics or methods mentioned in this report are not clear or problems occur with computation, users should seek the help of expert survey statisticians who are familiar with these methods. The software developers for the SUDAAN and WesVarPC can usually be contacted for further assistance with specific applications. ***Again, please consider that (1) sampling weights should be used in estimation to account for sampling variability and to adjust for differential probability of selection of persons in such a complex sample, and (2) the survey design structure should be used to estimate measures of statistical confidence.***

## APPENDIX A

**Table A.1: 1980 Census Population by age groups**

| Age groups    | U.S. population       |                           |             |
|---------------|-----------------------|---------------------------|-------------|
|               | Proportion<br>(total) | Proportion<br>(20+ years) | Total       |
| Under 1 year  | 0.0156                |                           | 3,533,692   |
| 1 - 2 years   | 0.0287                |                           | 6,493,373   |
| 3 - 5 years   | 0.0419                |                           | 9,483,880   |
| 6 - 11 years  | 0.0920                |                           | 20,834,439  |
| 12 - 19 years | 0.1418                |                           | 32,113,079  |
| 20 - 29 years | 0.1803                | 0.2650                    | 40,839,623  |
| 30 - 39 years | 0.1392                | 0.2046                    | 31,526,222  |
| 40 - 49 years | 0.1005                | 0.1477                    | 22,759,163  |
| 50 - 59 years | 0.1030                | 0.1514                    | 23,325,286  |
| 60 - 69 years | 0.0833                | 0.1225                    | 18,870,102  |
| 70 - 79 years | 0.0512                | 0.0752                    | 11,591,846  |
| 80 years plus | 0.0228                | 0.0336                    | 5,175,100   |
| Total         |                       |                           | 226,545,805 |

Source: 1980 civilian noninstitutionalized population of the U.S., U.S. Bureau of the Census, U.S. Department of Commerce.

**Table A.2: 1980 Census population by single year of age**

| Age          | Total population | Age      | Total population | Age        | Total population |
|--------------|------------------|----------|------------------|------------|------------------|
| Under 1 year | 3,533,692        | 40 years | 2,468,083        | 80 years   | 723,049          |
| 1 years      | 3,269,557        | 41 years | 2,375,849        | 81 years   | 640,276          |
| 2 years      | 3,223,816        | 42 years | 2,325,572        | 82 years   | 566,548          |
| 3 years      | 3,179,441        | 43 years | 2,237,108        | 83 years   | 527,982          |
| 4 years      | 3,141,748        | 44 years | 2,262,796        | 84 years   | 477,178          |
| 5 years      | 3,162,691        | 45 years | 2,242,318        | 85 years   | 412,549          |
| 6 years      | 3,109,095        | 46 years | 2,139,385        | 86 years   | 350,655          |
| 7 years      | 3,273,052        | 47 years | 2,222,969        | 87 years   | 306,906          |
| 8 years      | 3,394,998        | 48 years | 2,163,709        | 88 years   | 236,314          |
| 9 years      | 3,760,120        | 49 years | 2,321,374        | 89 years   | 213,778          |
| 10 years     | 3,716,530        | 50 years | 2,347,068        | 90 years   | 175,900          |
| 11 years     | 3,580,644        | 51 years | 2,295,077        | 91 years   | 140,003          |
| 12 years     | 3,518,982        | 52 years | 2,363,152        | 92 years   | 101,492          |
| 13 years     | 3,643,189        | 53 years | 2,337,138        | 93 years   | 78,233           |
| 14 years     | 3,782,784        | 54 years | 2,367,597        | 94 years   | 60,964           |
| 15 years     | 4,059,898        | 55 years | 2,390,440        | 95 years   | 46,219           |
| 16 years     | 4,180,875        | 56 years | 2,329,790        | 96 years   | 32,789           |
| 17 years     | 4,223,848        | 57 years | 2,312,737        | 97 years   | 23,471           |
| 18 years     | 4,251,779        | 58 years | 2,330,373        | 98 years   | 16,215           |
| 19 years     | 4,451,724        | 59 years | 2,251,914        | 99 years   | 12,385           |
| 20 years     | 4,387,100        | 60 years | 2,160,937        | 100 years  | 9,663            |
| 21 years     | 4,285,763        | 61 years | 2,073,764        | 101 years  | 5,231            |
| 22 years     | 4,284,351        | 62 years | 2,008,093        | 102 years  | 3,886            |
| 23 years     | 4,199,711        | 63 years | 1,931,425        | 103 years  | 2,800            |
| 24 years     | 4,161,779        | 64 years | 1,913,402        | 104 years  | 2,015            |
| 25 years     | 4,116,218        | 65 years | 1,904,641        | 105 years  | 1,573            |
| 26 years     | 3,977,515        | 66 years | 1,813,987        | 106 years  | 1,276            |
| 27 years     | 3,931,620        | 67 years | 1,763,637        | 107 years  | 1,038            |
| 28 years     | 3,708,968        | 68 years | 1,678,740        | 108 years  | 883              |
| 29 years     | 3,786,598        | 69 years | 1,621,476        | 109 years  | 852              |
| 30 years     | 3,726,525        | 70 years | 1,516,900        | 110 years  | 819              |
| 31 years     | 3,607,610        | 71 years | 1,439,723        | 111 years  | 623              |
| 32 years     | 3,712,217        | 72 years | 1,371,235        | 112+ years | 1,535            |
| 33 years     | 3,653,921        | 73 years | 1,261,994        |            |                  |
| 34 years     | 2,860,647        | 74 years | 1,208,272        |            |                  |
| 35 years     | 2,902,331        | 75 years | 1,111,480        |            |                  |
| 36 years     | 2,929,040        | 76 years | 1,028,927        |            |                  |
| 37 years     | 2,982,533        | 77 years | 951,774          |            |                  |
| 38 years     | 2,598,636        | 78 years | 828,866          |            |                  |
| 39 years     | 2,552,762        | 79 years | 872,675          | Total      | 226,545,805      |

Source: 1980 civilian noninstitutionalized population of the U.S., U.S. Bureau of the Census, U.S. Department of Commerce.

**Table A.3: Overall interview and examination response rates, NHANES III, 1988-94**

| Status                    | Sample size | Percent | Weighted Percent |
|---------------------------|-------------|---------|------------------|
| Total                     | 39695       | 100.0   | 100.0            |
| Not interviewed           | 5701        | 14.4    | 18.2             |
| Interviewed, not examined | 2683        | 6.8     | 7.5              |
| MEC examined              | 30818       | 77.6    | 73.4             |
| Home examined             | 493         | 1.2     | 0.8              |

Source: The NHANES III data file, 1988-94